MIL-C-48083A (AR)
19 August 1981
SUPERSEDING
MIL-C-48083
1 September 1972

MILITARY SPECIFICATION

CONTAINER, CARTRIDGE FOR 4.2 INCH AMMUNITION

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not covered by the drawings and provides quality assurance provisions for the fabrication of parts of one type of cartridge container for the Cartridge, 4.2 Inch, HE, M329A2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS (See 6.4)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

9245522 - Container, Cartridge

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with applicable drawings and specifications.

3.2 Parts. The parts shall comply with all requirements specified on drawing (dwg.) 9245522 and with all requirements specified in applicable specifications and standards.

3.3 Macroetch test (for metal defects). Parts shall be free from cracks, splits, cold shuts, inclusions, porosity, laminations or similar defects when tested as specified in 4.4.1.

3.4 Microetch test (for heat treatment). The aluminum stock (prior to machining) shall not be over-heated during heat treatment. Evidence of over-heating shall be indicated by the presence of rosettes in the microstructure or incipient melting at the grain boundaries when tested as specified in 4.4.2.

3.5 Workmanship. All parts and assemblies shall be free of burrs, chips, grease, rust, dirt, corrosion products and other foreign matter. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

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4.2 Classification of inspection. Quality conformance inspection shall be conducted on this item.

4.3 First article inspection. Not required

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, the extrusion lot shall consist of not more than 1000 pounds of extruded stock of the same alloy that has been thermally treated by the same method in the same furnace equipment at the same temperature and time in one unchanged process regardless of the solution heat treatment method utilized. Each extrusion lot representing material that has been solution heat treated after extrusion, must be solution heat treated at the same time in the same furnace using the same temperature and time in that furnace. The extrusion lot thus constituted shall be the basis for selecting the samples for the microstructure herein after prescribed.

4.4.2 Examination. See MIL-A-48078. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48083A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Bar, rod or tubing				NEXT HIGHER ASSEMBLY 9245522
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Macroetch test		100%	3.3	4.5.1
102	Microetch test	*		3.4	4.5.2

NOTES: * One sample from one end of one of every 100 bars shall be utilized for this test. Failure of a sample to meet the microetch requirements shall be cause for rejection of the heat treat batch.

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48083A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 9245522
4.4.2.2	Container, cartridge				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Number of flash holes improper		100%		Visual
102	Pitch diameter of internal thread		0.40%		Gage
103	Minor diameter of internal thread, max.		0.40%		Gage
104	Pitch diameter of external thread		0.40%		Gage
105	Major diameter of external thread, min.		0.40%		Gage
106	Runout of external thread pitch diameter with internal thread pitch diameter		0.40%		Gage
107	Length of external thread, min.		0.40%		Gage
108	Length of internal thread, min.		0.40%		Gage
109	Diameter of flash holes		0.40%		Gage
110	Burrs in cavity		0.40%		Visual
111	Burrs on outside diameter		0.40%		Visual
Minor					
201	Total length		0.65%		Gage
202	Outside diameter		0.65%		Gage
203	Inside diameter		0.65%		Gage
204	Width of propellant holder slot		0.65%		Gage
205	Diameter of propellant holder slot		0.65%		Gage
206	Length from internal thread end face to groove		0.65%		Gage
207	Length from external thread end face to first shoulder		0.65%		Gage
208	Length between shoulders external thread end		0.65%		Gage
209	Diameter of step external thread end		0.65%		Gage

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48083A (AR)

PARAGRAPH	TITLE		SHEET		DRAWING NUMBER
4.4.2.2	Container, cartridge		2 OF 2		9245522
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
210	Location of flash holes		0.65%		Gage
211	Depth of c'bore, max.		0.65%		Gage
212	Protective finish improper or incomplete		0.65%		Visual
213	Evidence of poor workmanship		0.65%		Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.4.3 Testing. Testing is described in the Quality Conformance Inspection tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of Inspection.

4.5.1 Macroetch test. Full section macroetch specimens shall be cut from the front and back end of the bar representing the first and last extruded material, respectively, from each individual extrusion billet. Failure to comply with the requirements of 3.3 will require an additional length of discard from the defective extrusion end(s) and testing of additional macroetch specimen(s) taken from the end(s) of the remaining length of extruded material. This procedure must be continued until a sufficient length of the defective extrusion has been discarded and the requirements of 3.3 are met. Suitable identification of each length of extrusion and corresponding macroetch must be maintained until the extruded material is accepted. The specimens shall be macroetched as specified in ASTM Method E-340 except that sodium hydroxide shall be used as the macroetch reagent.

4.5.2 Microetch test. Full section microetch specimen shall be cut from one end of a bar representing the extrusion lot (heat treat batch). When suitably prepared, the specimen shall be etched with Keller's reagent and examined by scanning under a metallurgical microscope at a minimum magnification of 500 diameters. Over-heating shall be evidenced by the presence of rosettes in the microstructure or incipient melting at the grain boundaries. Evidence of the above shall be cause for rejection of the heat treated extrusion lot.

5. PREPARATION FOR DELIVERY

5.1 Packaging and marking. Packing and marking shall be in accordance with Standard MIL-STD-1169.

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6. NOTES

6.1 Intended use. The components covered by this specification are intended for use on the 4.2 Inch, HE, M329A2 Cartridge.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment for design approval. See MIL-A-48078. Submit designs as required to Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAT-I, Dover, NJ 07801.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

Custodian:
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Preparing Activity:
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